

including, for example, the user device 110, multiple servers, or the like.

[0029] As illustrated in FIG. 4, the method 400 includes generating, with the electronic processor 300 (executing the data graph creation software 315), a data graph for an entity (at block 405). The data graph may be a provenance chain that represents the history or journey of the entity, such as a production history, distribution history, or the like. As noted above, the entity may include a physical product such as a car, a digital entity, such as a music file or social media news article, financial transaction, a person's medical history, a computer software product, or the like.

[0030] The process of generating a data graph, such as a provenance chain, can be described as "walking the tree." For example, for purposes of explanation, a data graph, such as a provenance chain, can be described and viewed as including a plurality of connected nodes organized in a tree structure, wherein each of the plurality of nodes represents a stage of an entity, such as a stage of production or distribution, a location, an input or sub-component, or the like. Connections between any two of the plurality of nodes represent connected stages and may represent changes in stages over a period of time, location, or the like. Accordingly, the resulting or final entity in the data graph may represent a root of the tree with other nodes representing branches or leaves in the tree. Also, each node in the tree includes at least one feature, such as a time (date, time, or combination thereof), a stage identifier, an entity identifier, or the like. For example, FIG. 5 illustrates an example data graph 500 that may be created by the electronic processor 300 at block 405. As illustrated in FIG. 5, an entity (product identified by package number 2176976) represents one node on the tree and other nodes represent components or stages that went into the processing, assembling, exporting, manufacturing, and packaging of the entity. It should be understood that a provenance chain can represent a historical path for a single entity or a group of entities that share a common path.

[0031] Accordingly, to "walk the tree" and create a data graph, such as a provenance chain for an entity, the electronic processor 300 accesses a plurality of data records stored in a first set of the plurality of data sources 115, wherein each of the plurality of data records is explicitly linked to the entity. In some embodiments, the first set of the plurality of data sources 115 includes one or more data sources provided by one or more organizations or individuals, such as, for example, organizations and individuals associated with a plurality of stages (for example, the production, distribution, or both) of the entity. As used herein, explicitly linked to an entity means that the data record is linked (directly or indirectly) to an identifier of the entity as described below.